

How much does the map of science depend on the embedding model?

Alejandro Treny Ortega

University College London, London, United Kingdom

21 September 2026

Abstract

Science maps describe relationships between papers. Text embedding models build these relationships by turning a paper’s text into a list of numbers that gives it a position. Would another model give the same picture? I compare ten models using titles and abstracts of 500,000 publications from the OpenAlex bibliographic catalogue, published in 2000–2024 across 26 disciplines, including Medicine and Physics. Representing each discipline by its papers’ average position produces similar arrangements across models, including when other papers are selected. Yet within a discipline and period, two models share, on average, only eight of the 25 papers closest to a given paper. Supplying titles alone also changes much of that list, with effects that vary between models. Models can also disagree about which discipline’s papers are more spread out. One measure of spread compares angles between lines drawn from a common point to each paper’s position. Moving that point can reverse which discipline appears more spread out, even when selecting other papers leaves the answer unchanged. The broad arrangement can remain similar while answers about nearby papers and differences between disciplines depend on the model, the text it receives and how its output is processed.

Keywords: science mapping; document embeddings; robustness; neighbourhood agreement; geometric comparisons.

1 Introduction

A science map can help a reader find related papers or compare research areas. Its usefulness depends on the relationships it shows. When a model determines those relationships from text, choosing the model becomes part of choosing the map. Two models may lead a reader to different papers near the same starting article, or disagree about how widely an area’s papers are spread. I ask how much the answer changes when we keep the papers and use another model to represent them.

The models studied here read a paper’s title and abstract and turn that text into a list of numbers, called a vector or *embedding*. The numbers act as coordinates: together they give

the paper a position from which distances to other papers can be calculated. Papers placed close together are treated as related. This approach has been used to map biomedical literature and the structure of physics (González-Márquez et al., 2024; Constantino et al., 2025). But the position comes from the model. It is not a relationship observed independently of it, and the title and abstract do not contain everything in the paper.

This concern has a history in science mapping. Gläser et al. (2017) asked how much a reconstructed thematic structure reflects scientific knowledge and how much it reflects the method used to recover it. Boyack and Klavans (2020) found that methods could obtain similar overall accuracy while assigning different papers to clusters. An overall score can therefore leave an important question unanswered: would another method support the particular comparison a reader wants to make?

I use ten models and a fixed collection of 500,000 publications to examine three questions. The first concerns shared relationships: do models preserve the broad arrangement of areas and identify the same nearby papers? The second asks how much of a change comes from what the model reads: does supplying only a title matter as much as switching models? The third, added as a later exploration, concerns a description we might draw from the map: do models agree that one area’s papers are more spread out than another’s? Repeating each type of comparison with other selected articles also shows whether its answer depends on the particular papers used.

The contribution is to follow the comparisons that a reader might draw from a science map, and to test which choices they survive. A claim that one area’s papers are more spread out than another’s can repeat when different papers are selected, yet receive opposite answers from different models. I examine these persistent disagreements alongside the size of the differences and their dependence on processing choices. This connects a broad assessment of model agreement to the reliability of particular comparisons between research areas.

2 Background and research questions

2.1 What earlier comparisons tell us

Previous studies assess scientific embeddings against several useful criteria. Constantino et al. (2025) compare them with disciplinary classifications in physics; González-Márquez et al. (2024) examine related biomedical papers and their labels. Such checks ask whether a representation agrees with information about the literature. The present study asks a complementary question: which relationships remain when the representation changes?

The model is only one choice involved. The *input text* is the text we give it to read: a title alone conveys less information than a title followed by its abstract. Models also accept different lengths of text and use different rules to produce a single vector for it. Singh and

Singh (2022) show that scholarly search can be sensitive to the wording of short queries. The supplied text therefore belongs in the comparison alongside the model.

Several parts of this comparison have direct precedents. Caspari et al. (2024) combine structural similarity with overlap in retrieved documents. Imel and Hafen (2025) examine local geometry in scientific literature, while Raju (2026) distinguishes representational similarity from the stability of geometry. Here, I bring these concerns to specific comparisons between research areas. For each pair of areas, I ask whether a model keeps the same ordering across article selections, whether another model persistently supports the opposite ordering, and whether those same models retain their answers after a change in processing. Matched text and search comparisons then help establish which other choices affect the result.

2.2 What agreement can establish

Two changes need to be kept separate. To measure *agreement*, I keep the articles fixed and compare models. To check *selection stability*, I keep the model fixed and choose other articles under the same rules. An *article selection* is simply a set chosen from the existing collection; repeating it does not mean downloading new papers. A result can repeat with other articles in each model while the models consistently give opposite answers.

Neither property establishes that the answer is correct about the research itself. Models can share an error, and a repeatable pattern can be hard to interpret. In a study of invasion biology, Held and Velden (2022) show why topic solutions require substantive interpretation beyond numerical checks. Here, agreement is evidence that a result survives a change of model, not evidence of thematic truth.

The word map needs one further distinction here. I compare the numerical positions, each with hundreds of coordinates, before reducing them to a drawing. A two-dimensional plot would add a transformation that could change apparent distances. *Structure* means the pattern of relationships among the objects being compared, whether papers or area summaries. *Spread* describes how far apart the representations lie within one area. Two models can share much of the broad structure without agreeing on every area's spread.

2.3 Research questions

RQ1. When the model changes, what remains of the arrangement of research areas and of the papers placed close together?

RQ2. How much does changing the input text, or the rule used to form one paper vector, matter compared with changing the model?

RQ3. Do models agree about which area’s paper representations are more spread out, or whether that spread is concentrated along a few directions rather than many? Do these comparisons repeat with other articles?

The third question was added after the initial comparisons and is exploratory. All three concern a fixed collection and a fixed set of models. They do not seek a winning model or estimate the structure of all scientific literature.

3 Data and methods

3.1 One collection, several questions

OpenAlex is a bibliographic catalogue that supplies titles, abstracts and other information about publications (Priem et al., 2022). The starting collection, or corpus, contains 500,000 distinct work IDs retrieved and cleaned by 15 September 2026. Records were labelled as English articles, reviews or conference papers published in 2000–2024. Each required a title, an abstract of at least 50 words reconstructed from OpenAlex’s stored format, and a primary research area. Retractions, ancillary material and clear content failures were excluded. Less certain cases and possible duplicate versions were retained with review flags. Those flags record a concern for later checks; they do not establish that every retained record is correct. Supplement S1 gives the rules and counts.

OpenAlex assigns papers to broad *Fields* and narrower *Subfields*, using information from text, citations and journals (OpenAlex, 2026). I call these areas and specialties. For example, Medicine is an area; Cardiology and Cardiovascular Medicine is one of its specialties. The 26 areas provide the main units of comparison. These database labels organise the analysis; they are not independently verified descriptions of every paper.

The collection has two parts. Reproducible random requests to OpenAlex supplied 400,000 eligible records. Another 100,000 were added to strengthen groups that initially had few papers from the same area and five-year period. For example, Medicine in 2000–04 is one such group. The five periods are 2000–04, 2005–09, 2010–14, 2015–19 and 2020–24. Combined with the 26 areas, they give 130 area–period groups, each with at least 2,165 papers. Searches across the whole general collection use the original 400,000 records; comparisons within areas can use both parts. The reinforcement makes smaller groups easier to compare, but changes

their share of the collection. Neither that share nor an average giving every area equal weight estimates world publication proportions.

All ten models represent the 500,000 papers using titles and abstracts, but each subsequent question uses the articles it needs. The *input-text experiment* compares titles, abstracts and both together on the same 52,000 papers: 2,000 per area, with 400 from each period. These include the 26,000 used in an initial trial. The expansion tests how the agreement measures and their sensitivity to article selection change as more papers are included. The later study of spread within areas reuses this set and its title-plus-abstract vectors. The *common-passage check* uses a different set of 52,000 papers to ensure that every model reads exactly the same passage, short enough to fit all input limits. This separates a choice about content from a choice about length. Table 1 is a guide to these comparisons; its rows overlap and should not be added together. Within any comparison, the models always represent the same paper IDs.

3.2 How each model represents a paper

The ten models in Table 2 were already trained before this study. I use their published versions to convert text into vectors, without training them again on this collection. They cover scientific documents, general sentence similarity, and scientific or biomedical language. I prioritised models used most often in the science-mapping studies reviewed, such as SPECTER and SciBERT. I added models used as comparators in those studies to cover different training approaches (González-Márquez et al., 2024; Constantino et al., 2025; Liang et al., 2026). This is a fixed comparison set, not a random sample of models or a ranking of the newest systems. Versions, input formats and software settings were fixed before calculating the vectors.

The main input is the title followed by the abstract, in each model’s usual format. Models divide text into small pieces called *tokens*, which can be whole words or parts of words. A token limit is therefore not a word limit. Under their standard settings, MiniLM reads at most 256 tokens and MPNet 384; text beyond that limit is cut off. This affects 49.58% and 20.16% of the papers, respectively. I retain those settings for the main comparison. Because MiniLM cuts off text in almost half the papers, I also test it separately at 512 tokens to check whether reading more text changes its agreement with the other models. The common-passage check addresses the resulting difference in what models can read: all receive the same content, although each splits it into tokens in its own way.

A further choice concerns how a model produces one vector for the whole paper. BERT, SciBERT, BioBERT and PubMedBERT produce an output vector at each token position, but were not trained specifically to compare whole documents. The main rule averages those outputs after the model has finished processing the text. This step is called *mean pooling*. The average includes special symbols used to mark the text and excludes empty

Table 1: Questions, comparisons and article sets.

Question	What is compared	Articles used
Does the broad arrangement stay similar?	Ten models: relationships between groups, summarised by their average positions, and between papers within groups.	256 papers per group to calculate its centre; repeated choices and size checks at 128 and 512.
Do models find the same nearby papers?	How many of the 25 closest papers appear in both models' lists.	All papers within each area and period; a separate check fixes 2,048 candidates per group. Global search: 13,000 starting papers among 400,000 candidates.
Does changing the input text matter?	The same model reads only the title, only the abstract, or both.	52,000: 2,000 per area, 400 per period. Includes an initial 26,000-paper trial.
Does reading the same passage matter?	Each model's usual input versus a common passage that fits every model.	A different set of 52,000, with 400 per area and period.
Do models agree on how papers spread within areas?	Angles between paper vectors and how their variation spreads across directions; all 325 area pairs under ten models.	The same 52,000 as in the input-text experiment; further article choices test whether the comparisons repeat.
Does searching a narrower specialty matter?	For example, Cardiology and Cardiovascular Medicine versus Medicine, with 256 candidates in each search.	50 fixed starting papers per specialty; ten choices of the other 206 candidates. In total, 10,850 starting papers across 217 specialties.

All sets come from the same 500,000-paper collection: 400,000 general records plus 100,000 added to strengthen small area-period groups. Rows are not added together. Models use the same paper IDs within each comparison. Candidates are the papers available for a neighbour search. In the scope comparison, both candidate sets include the same 50 starting papers. Methods and Supplements S1–S4 and S9 give the full controls. The 1,300 software-test papers are not another scientific sample.

positions added as padding. Two alternatives take only the output at the first special-symbol position (CLS) or at the final non-padding position (SEP). These outputs have processed the surrounding text; they do not represent only the first or last word. The mean rule was specified before the primary comparisons and retained after early checks, rather than chosen to maximise agreement. The other six models use their documented rules. Table 2 and Supplement S1 specify the exact settings.

Table 2: Embedding models and primary representation settings.

Model	Vector size	Token limit	Paper vector	Reference
SPECTER	768	512	CLS	(Cohan et al., 2020)
SPECTER2	768	512	CLS + adapter	(Singh et al., 2023)
SciNCL	768	512	CLS	(Ostendorff et al., 2022)
SciBERT	768	512	Mean	(Beltagy et al., 2019)
BERT	768	512	Mean	(Devlin et al., 2019)
MPNet	768	384	Mean	(Reimers and Gurevych, 2019)
MiniLM	384	256	Mean	(Reimers and Gurevych, 2019)
PubMedBERT / BiomedBERT	768	512	Mean	(Gu et al., 2022)
BioBERT	768	512	Mean	(Lee et al., 2020)
SimCSE	768	512	CLS	(Gao et al., 2021)

Vector size is the number of coordinates, not the number of words read. MPNet and MiniLM use all-mpnet-base-v2 and all-MiniLM-L6-v2, respectively. Limits count tokens, the pieces into which a model divides text, including its special symbols. Mean averages the final token outputs (the last layer), including special symbols and excluding empty padding positions. CLS uses the first-position output before the additional transformation called the pooler. SPECTER2 also uses its proximity adapter, an additional trained component. For MPNet/MiniLM, the reference describes the Sentence-BERT framework, not the exact training of these published versions. Changing the pooling rule does not add a separately trained model. Full revisions, formats and truncation: Table S2.

3.3 Following the map from areas to individual papers

A vector can be pictured as an arrow from a common starting point to the paper’s position. Before comparison, I scale every arrow to length one, keeping the direction in which it points. This operation is called *normalisation*; it prevents differences in vector length from determining closeness. To summarise an area, I average its article vectors and also scale the resulting vector to length one. I call this summary its *centre*. Comparing the centres asks whether models place areas in similar relationships to one another. Comparing the individual papers inside an area asks how much of its internal organisation they share.

The first centre comparison fixes 256 papers per group, so that unequal group sizes do not drive the comparison. It considers the 26 areas, the 217 specialties, and a restricted set containing one specialty from each area. To test dependence on which papers were chosen, I repeat article selection fifty times while keeping each group’s counts from the five publication periods. To test dependence on size, I use 128, 256 and 512 papers for the same 26 areas and

the 183 specialties that have enough papers at every size. These are comparison settings, not a claim that one size is universally sufficient.

Each selection also supplies a reference: the same papers are randomly redistributed into groups while preserving group sizes and publication-period counts. The reference asks how much model agreement remains when groups no longer follow the area or specialty labels. Further checks leave out one area at a time and, for the restricted specialty set, separate choosing a specialty from choosing its papers. Supplement S9 details these centre checks.

3.4 Finding and comparing nearby papers

For an individual paper, its *neighbours* are the papers placed closest to it. The *starting paper* is the one whose neighbours we seek; the *candidates* are all papers available for that search. Each model returns the closest 25, and agreement is the fraction that appears in both models' lists. Lists of 10 and 50 provide checks. Closeness uses cosine similarity: arrows pointing in more similar directions count as closer. The starting paper cannot be its own neighbour. The lists are calculated exactly and checked by independent sorting.

A search depends on the papers available to it. Searching among every paper from the same area and period gives different groups different numbers of candidates. A separate check fixes 2,048 candidates per group to remove that difference in count. Within a group, every model searches the same papers. A further search uses the 400,000 general records without area or period restrictions. These comparisons answer different questions, so each result specifies its candidate set.

The quality check asks whether flagged records and duplicates explain disagreement between models. Applying the existing review flags and duplicate rule leaves 448,886 records. I compare neighbour lists for the same retained starting papers before and after filtering the candidates. Removing records also reduces search size, so further checks hold that size fixed: 1,024 candidates in all 130 groups, and 2,048 in the 63 groups still large enough after filtering. The before-and-after sets include the same 100 clean starting papers per group. Here, clean means passing the recorded filter; it does not mean that every possible source error has been ruled out.

To separate search scope from search size, I compare a specialty with its parent area using the same candidate count and matched dates. Consider 50 fixed starting papers from Cardiology and Cardiovascular Medicine. For each, we search among 256 cardiology papers and among 256 papers from Medicine. Both sets include those same 50 starting papers, leaving 206 places for others; the current starting paper is excluded when its neighbours are ranked. I choose the other 206 ten times, keeping the 50 fixed and matching publication-period counts between searches. The procedure covers 217 specialties, giving 10,850 starting papers. It deliberately includes the 50 specialty papers in the broader set too. Its result therefore

applies to this matched comparison, rather than to an unrestricted random selection from Medicine.

3.5 Measuring shared structure

To measure shared structure, I compare the pattern of relationships among the same objects under two models. The objects can be individual papers or the centres that summarise areas. The main measure is corrected linear centred kernel alignment, abbreviated CKA (Kornblith et al., 2019; Murphy et al., 2024). It compares these patterns without requiring a coordinate in one model to mean the same thing as a coordinate in another. A value near one, such as 0.9, indicates that the patterns are very similar on this measure. It does not mean that a corresponding percentage of science is correctly represented. The correction reduces a known upward distortion associated with finite samples; it does not remove all statistical bias.

Two further measures check other aspects of shared structure. Procrustes alignment allows the representations to rotate and change scale before comparing them. The distance-rank check asks whether pairs of papers have a similar ordering from nearer to further apart. These measures need not give identical answers (Williams et al., 2021; Williams, 2024). Their formulas appear in Supplement S2.

3.6 Describing spread within an area

A different question concerns the shape formed by the papers within each area. I describe two properties. *Angular spread* summarises the angles between their vectors, using the median-distance convention in Supplement S6. Picture arrows drawn from a common point to the papers: larger angles mean that the arrows point further apart.

Effective linear dimension describes how that cloud of paper positions spreads across directions. Here, variation means differences between the papers' coordinates. A cloud concentrated along a line varies mainly in one direction; one spread across a plane varies in more. The participation ratio, or PR, extends this idea to the full vector space and allows values between whole numbers. A larger PR means that variation is less concentrated in a few directions. Neither PR nor angular spread counts research topics or establishes thematic diversity.

Alternative angle summaries and dimension measures check how much the answer depends on the formula. For dimension, the entropy-based alternative measures how evenly variation is distributed. D80 counts the fewest directions needed to account for 80% of that variation. Because these alternatives describe the same underlying variation, agreement among them would not independently validate a thematic interpretation. I also tested a further idea: whether weak links between nearby papers could reveal an area split into separate groups.

Artificial examples exposed problems with reading that signal as thematic fragmentation. Supplement S6 gives the definitions and controls.

3.7 Checking whether a comparison repeats

With 26 areas, there are 325 ways to choose two different areas, counting each pair once. For each property, each model gives a value for area A and a value for area B. I ask whether the same area keeps the larger value when different papers are chosen. The main calculation uses 2,000 papers per area. I repeat it with twenty half samples of 1,000 and five further selections of 2,000. An ordering is *persistent* only if it holds in all 26 conditions. This refers to the order of the two measured values, not to the direction in which a vector points. The dimension alternatives use the same conditions.

Another check changes the common point from which the arrows are drawn. For each model, I calculate the mean of the balanced 52,000 unit vectors used in the input-text experiment, subtract it from every vector, and scale each resulting vector back to length one. This is *global centring followed by normalisation*. The new origin is the average position in this set. I then recalculate spread and dimension to ask whether comparisons between areas depend on that choice. Supplement S6 distinguishes this change to the vectors from the internal centring already used in the PR formula.

The results fall into three classes. In *unanimous agreement*, all ten models persistently give the same area the larger value. In *opposition*, at least two fixed models persistently give opposite orderings. Otherwise the comparison is *unresolved*. For example, some models may change their ordering with other articles and no fixed pair may maintain opposite answers. Unresolved does not mean equal. Nor does a tie in one model necessarily leave the whole comparison unresolved: two other models can still maintain opposite orderings.

I also ask whether the differences are large enough to matter numerically. Besides any nonzero difference, I require minimum differences of 1, 5 and 10% of the two areas' average value. A model's ordering counts at a given cutoff only if its difference exceeds that cutoff in every selection. The three classification rules are then applied again. These percentages describe the size of a measured difference, distinct from the percentage of the 325 pairs assigned to a class. They are sensitivity checks, not universal thresholds of scientific importance.

A separate diagnostic, the *selection alert*, marks a result that changes more than a recorded tolerance allows when we choose other articles or use another tested size. It identifies a result worth examining for sensitivity to article choice; it is neither a source-record error nor one of the three area-pair classes above. Each check has its own rule and count, retained in the supplement.

Selections overlap because they come from the same fixed collection. Their ranges describe changes among the available papers, rather than confidence intervals for all scientific literature. A central 95% selection range leaves out the lowest 2.5% and highest 2.5% of the

recorded results. Comparisons also reuse models and areas, so I describe their patterns without treating them as independent observations in significance tests.

The work developed in stages. The initial structure and neighbour comparisons came before the expanded text experiment and the specialty searches. The geometric analysis was a later exploration, and the summary of 325 area pairs followed its first examples. Rules were recorded before the corresponding new calculations, with earlier results already known; some controls were added along the way. Supplement S2 documents this sequence. The full study was not registered in advance with an external service, so these later checks should not be read as predictions made before any results were seen.

4 Results

4.1 The broad arrangement is shared more than the detail

Models share a similar arrangement of area centres, and choosing other papers changes that conclusion little. Each mean combines the 45 ways of pairing the ten models. For the original article selection, corrected CKA averages 0.939 for area centres and 0.911 for specialty centres. Across fifty selections, the averages are 0.940 and 0.911, with central 95% selection ranges of 0.937–0.943 and 0.910–0.912 (Figure 1A). Randomly redistributed groups average 0.627 and 0.636. None of their selection means reaches the lowest observed selection mean. Thus, the labelled groups produce more shared structure than these random references.

The arrangement also changes little when centres use more papers. At 128, 256 and 512 papers, area-centre means are 0.938, 0.940 and 0.941; the same 183 eligible specialties give 0.913, 0.914 and 0.915. Leaving out each area in turn gives 0.933–0.946 in the original comparison, so no single area accounts for the high agreement. When only one specialty represents each area, the mean is 0.904. In this restricted comparison, changing which specialty represents an area matters more than changing its selected papers (Table S18).

Agreement is lower when we compare the individual papers inside a group. Mean CKA stays around 0.63 within areas and 0.64 within the same 183 specialties as the count rises from 128 to 512 (Figure 1B). That stable average hides individual comparisons that change more. Of the 8,235 combinations of a specialty and a model pair, 2,628 trigger the selection alert (31.9%): their agreement changes more than the recorded tolerance allows across article choices or tested sizes. A stable mean therefore does not make every specialty result stable. The contrast with centres also concerns different objects—group summaries versus individual papers—so it does not isolate the effect of scale alone.

The alternative structure measures broadly agree about which model pairs are more similar. Within each of the original 130 area–period groups, I order model pairs from less to more agreement and compare those orderings. Their mean rank correlations with corrected CKA

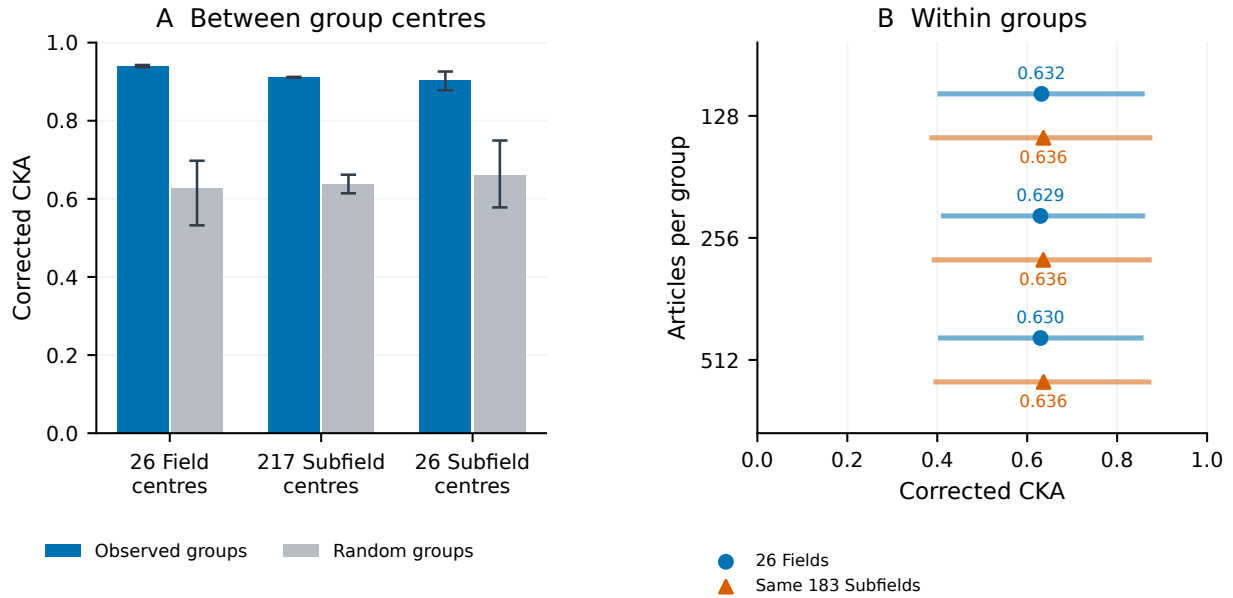


Figure 1: Agreement about group centres and individual papers. Higher corrected CKA means greater agreement between models. (A) Each centre summarises 256 papers. Area and specialty results average fifty article selections. The 26-specialty condition chooses one specialty from each of the 26 areas, repeats that choice fifty times, and chooses ten article sets within each choice. Random groups redistribute the same papers, preserving sizes and dates. Lines span the central 95% of selection means. (B) Agreement among individual papers within 26 areas and the same 183 specialties, at three article counts. Points show means; lines span the central 95% of group-model-pair results: 1,170 for areas and 8,235 for specialties. Panel A compares relationships between group summaries; panel B compares relationships between papers inside groups. Their ranges describe different sources of variation and are not confidence intervals.

are 0.891 for Procrustes and 0.881 for distance ranks; values nearer one mean more similar orderings. Agreement is not complete. The original article-selection check also retains 50 alerts among 5,850 comparisons, corresponding to 45 model pairs in each of the 130 groups (Table S4). This is a different check from the specialty alerts above, with its own sizes and tolerances.

4.2 Changing the model changes which papers appear nearby

When every paper in the same area and period is available as a candidate, two models share an average of 30.3% of the 25 nearest neighbours: about eight papers out of 25. The average gives equal weight to each area-period group and model pair. Holding every group's candidate count at 2,048 gives 31.9% agreement. The count is shared across groups; the actual papers are shared across models within each group, not across disciplines.

The general-collection search asks a broader question. It takes 100 starting papers from each of the 130 groups, giving 13,000 in total, and searches among the 400,000 general records. Agreement is 17.5%. This search both admits other disciplines and periods and offers many more candidates. Because both change together, the comparison cannot tell us how much of the lower agreement comes from each change.

Filtering source flags and duplicates changes little of this picture. To make the before-and-after comparison fair, both averages use only the starting papers that survive the filter. Their initial agreement is 30.58%, slightly different from the 30.3% for all starting papers above, and rises to 31.43% after filtering candidates. The gain is 0.85 percentage points. Both averages still mean roughly eight shared papers out of 25. Keeping candidate counts fixed reduces the gain to 0.28 points with 1,024 candidates and 0.19 with 2,048. The issues removed by this filter explain little of the overall disagreement, although some areas change more than others (Table S19).

The matched specialty searches isolate a change in disciplinary scope more closely: both searches have 256 candidates, the same starting papers and matched dates (Figure 2). Searching within the specialty makes models share fewer of their 25 neighbours in 127 of 217 specialties (58.5%), but more in the other 90 (41.5%). The average change from the parent-area search is -1.35 percentage points. With lists of 50 neighbours, the split is almost even: 109 specialties decrease and 108 increase. Narrowing the search therefore does not have the same effect everywhere. These results describe how much the models' lists overlap; they do not assess how useful the listed papers are.

Pairs of models that agree more on overall structure also tend to share more neighbours. Ordering the 45 model pairs by each type of agreement gives a rank correlation of 0.920. This association does not make the two checks interchangeable: a pair can preserve broad relationships while replacing many individual neighbours. The model pairs also share models with one another, so the association is descriptive rather than evidence from 45 independent observations.

Supplement S4 moves from averages to named papers: it checks whether a paper keeps the same neighbours when sample size, the rule for forming its vector, or neighbour-list length changes. Reading these examples also revealed source problems. A history book appears under Physics, a text about bacterial infection under Politics, and another record is an announcement. I checked these against the downloaded OpenAlex records: the classifications and content were already present there, rather than introduced by matching the wrong papers across models. The examples remain in the analysis. Because they were selected as cases to inspect, their number does not estimate the frequency of such problems in the collection.

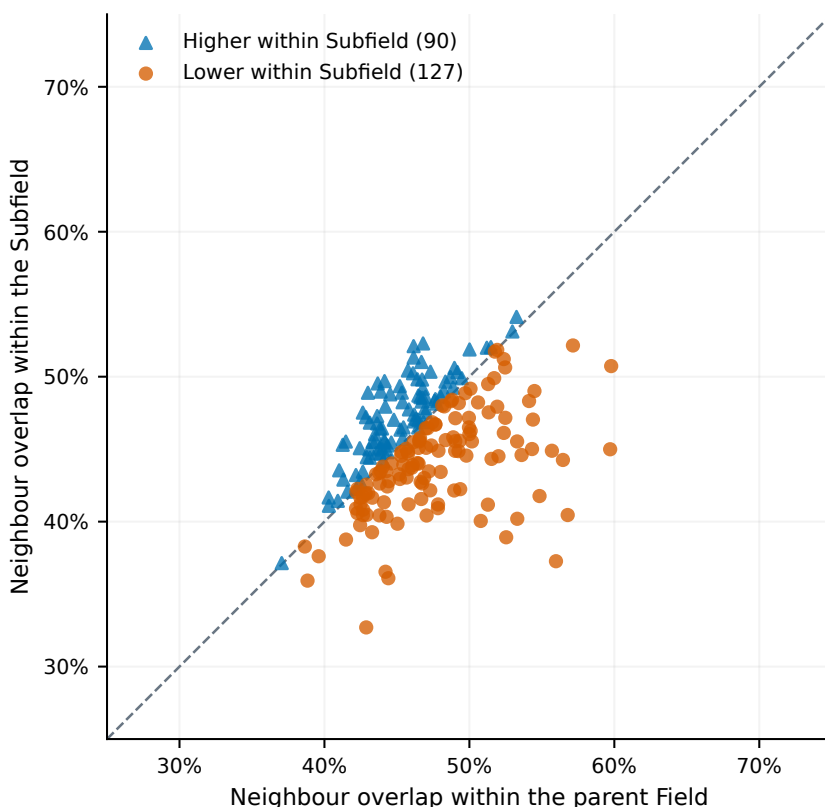


Figure 2: Does searching within a specialty change agreement? Each point represents one of 217 specialties. The horizontal axis shows the fraction of neighbours shared by models when searching the parent area, such as Medicine. The vertical axis shows that fraction within its specialty, such as Cardiology and Cardiovascular Medicine. Above the diagonal, models share more neighbours in the narrower search; below it, they share fewer. Both candidate sets contain 256 papers, including the same 50 starting papers, with matched dates. Values average model pairs and ten choices of the other 206 candidates. The parent-area set always includes those 50 specialty papers. Differences between points describe specialties, not confidence intervals.

4.3 Similar average changes hide different model behaviour

The input-text experiment now changes what the model reads. It uses the same 52,000 papers for three inputs: title alone, abstract alone, and title plus abstract. Here, each area contributes 2,000 candidates, combining its five periods. This differs from the preceding searches within a single area and period.

Starting from title plus abstract, I compare keeping the text and switching models with keeping the model and supplying only the title. The first replaces an average of 68.8% of the 25 neighbours; the second replaces 70.7% (Figure 3). These are fractions replaced, so a larger value means less agreement. The corresponding structural changes are 0.370 and 0.366, measured as one minus CKA: subtracting agreement from one makes larger values

mean more change here too. The two average changes are similar, but they need not be similar for every model.

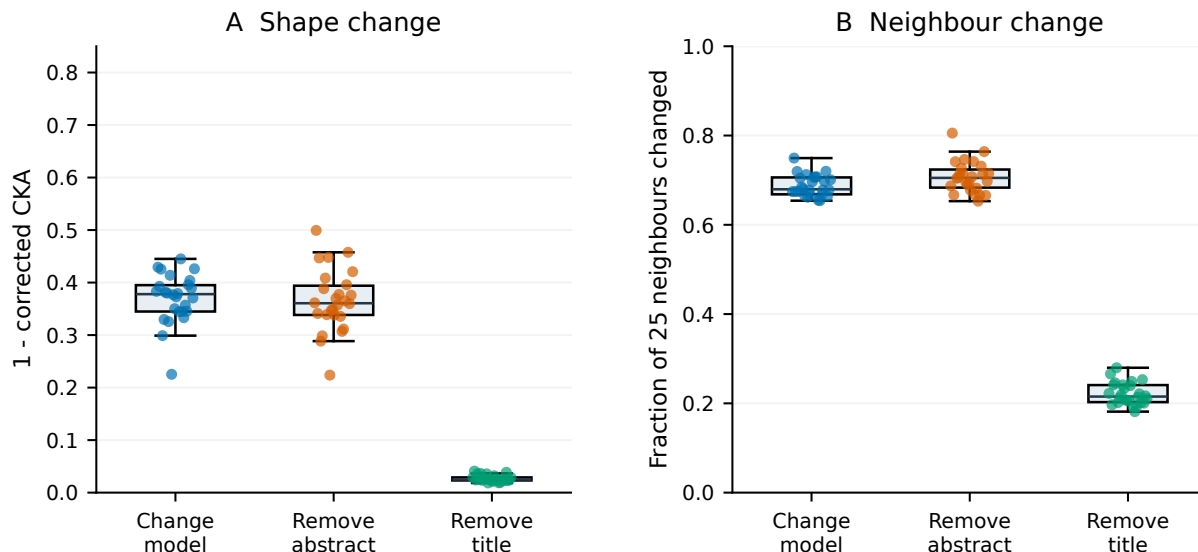


Figure 3: Changing the model or changing its input. All comparisons use the same 52,000 papers, with 2,000 per area across five periods. The starting input is title plus abstract. Model change compares models on that input; text change keeps the model and uses only the title or only the abstract. Larger values mean more change: (A) one minus corrected CKA; (B) the fraction of 25 neighbours replaced among the area’s 2,000 candidates. Each point is an area average. The line inside each box is the median; the box covers the middle half of area values. Whiskers extend to values within 1.5 box heights of the box edges. These distributions describe differences between areas, not confidence intervals. Models retain their usual text limits.

Keeping the abstract and removing only the title changes less: 0.027 for structure and 22.1% of neighbours. Titles alone also affect models differently, replacing 51.7% of neighbours for MPNet and 84.2% for SciBERT (Table S5). The similar overall averages therefore should not be read as a general equivalence between changing text and changing models.

To see whether that average comparison repeats, I choose fifty balanced sets of 1,000 candidates per area, half the full experiment’s count. These smaller searches replace 64.93% of the 25 neighbours when switching models and 66.49% with titles alone. Title-only input therefore replaces 1.56 percentage points more, with a central 95% selection range of 1.44–1.69. The overall difference stays positive in every repetition. It averages 1.97 points with 10 neighbours and 1.04 with 50.

The model-specific results tell a less uniform story. Differences range from -14.04 points for MPNet to $+16.59$ for SciBERT. A negative value means that titles alone change fewer

neighbours than switching models; a positive value means they change more. Thus, a repeatable positive grand mean combines models for which the two changes have opposite relative effects (Supplement S9).

Changing the rule that forms a paper vector can also reverse the overall result. For the four word-level models—BERT, SciBERT, BioBERT and PubMedBERT—I replace the mean of token outputs with the CLS or SEP output defined above. The other six models keep their usual rules. Both sides of the comparison are recalculated: the effect of switching models and the effect of supplying titles alone. With 25 neighbours, titles alone now replace 3.13 percentage points fewer for CLS and 2.00 fewer for SEP, in every repetition. The ordering therefore depends on this processing choice. These are comparisons of practical ways to use the models; they do not isolate the causal contribution of text, model design or a single processing step.

The expansion from the first 26,000 papers to 52,000 also reduced sensitivity to article selection in the structural comparison. Applying the same check with 100 repeated selections at each size lowers alerts from 72 of 520 model–area–input contrasts (13.8%) to five of 520 (1.0%). Fewer contrasts therefore exceed the recorded tolerance. This does not establish stability for every neighbour comparison. Supplement S3 retains the earlier twenty-selection check separately. The common-passage and MiniLM 512-token checks instead concern input length; they do not reveal a coding error in MiniLM’s original limit.

4.4 A comparison between two areas can reverse under another model

The final analysis asks whether models support the same description of an area. It reuses the input-text experiment’s 52,000 papers and their title-plus-abstract vectors. For example, do the arrows representing one area’s papers point further apart than those of another? Before comparing areas, I check whether each area’s measured value changes greatly when other papers are chosen.

Ten models and 26 areas give 260 cases to check. Angular spread has no selection alerts in either the half-sample or equal-size check. PR has four alerts (1.5%) with twenty half samples of 1,000 papers per area, and none with five further selections of 2,000. These checks use different sizes and repetition counts. Passing the second does not remove the sensitivity detected by the first.

Repeatable values within a model do not ensure agreement between models. Among the 325 area pairs, 80.6% have persistent opposite answers from at least two models for angular spread, and 68.0% for PR (Figure 4). For example, one fixed model gives area A a larger angular spread than B in every article selection, while another gives B the larger spread in every selection. The other models need not all disagree with one another. “Larger” refers to the measured property, not to a higher position in a drawing or better research.

Unanimous agreement is less common: all ten models persistently give the same area the larger value in 12.9% of pairs for angular spread and 16.6% for PR. The remaining 6.5% and 15.4% do not meet either the unanimous-agreement or opposition rule. They remain unresolved under those rules.

Many reversals concern small differences between areas. Requiring each persistent ordering to exceed a 5% difference, relative to the two measured values' average, reduces opposition to 29.5% of the 325 pairs for angular spread and 54.5% for PR. At a 10% minimum difference, the shares fall to 5.8% and 38.8%. The first percentage in each comparison is a requirement on the difference's size; the resulting percentage tells us how many pairs still meet it. A reversal over a tiny difference and one over a larger difference should not carry the same interpretation.

The alternative formulas often retain the disagreement, but not in every case. I count an opposition as retained only when at least one of the original opposing model pairs keeps the same area orderings. Alternative angle summaries retain 85.9% of the original angular oppositions by this rule (225/262). Across the same 26 article conditions, entropy-based dimension retains 94.6% of the original PR oppositions and D80 retains 86.0%; 86.0% survive both alternatives (190/221). These denominators are the original opposed pairs, not all area pairs. Because D80 returns whole numbers, it can give ties that contribute no persistent ordering.

The four word-level models do not account for all opposition either. Keeping only the six models trained for similarity still leaves opposite answers in 71.1% of all 325 pairs for angular spread and 49.2% for PR (Tables S14–S15).

Angular comparisons change much more when the common reference point moves to the mean position of the balanced 52,000 papers (Figure 4). After subtracting that mean and restoring unit vector length, opposition falls from 80.6% to 46.5% of all 325 pairs: from 262 to 151. But the new total does not tell us which original disagreements remain. Only 48.9% of the original oppositions retain at least one of the same opposing model pairs with the same area orderings (128/262). Table S20 separates these from new oppositions or those supported by different model pairs. With the 5% minimum-difference rule, only 11 opposed pairs remain instead of 96.

PR is less sensitive to this transformation. Opposed pairs fall from 221 to 211, and 92.3% of the original oppositions retain at least one of the same opposing model pairs and area orderings (204/221). An angular comparison that repeats with other papers can therefore still depend strongly on the origin from which angles are measured.

The attempt to identify separate groups from links between nearby papers reached a different limit. Weak connections might seem to suggest an area divided into research topics. Yet artificial elongated clouds and isolated extreme points produce that signal without any thematic groups. It also changes with sample size and the rule for linking neighbours, with selection alerts in 51 of 260 cases. Supplement S6 retains this unsuccessful check because it

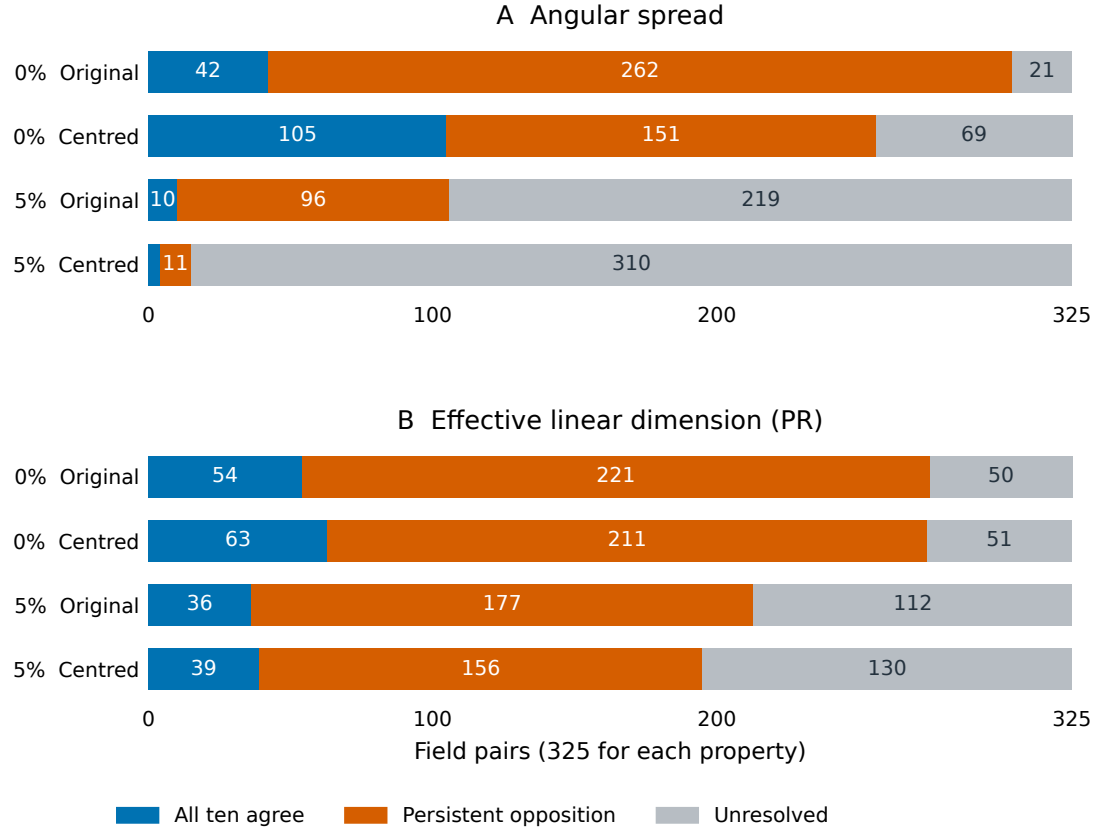


Figure 4: How model agreement changes with the representation. Each bar covers all 325 pairs of areas. Blue: all ten models persistently give the same area the larger measured value. Orange: at least two fixed models persistently give opposite orderings. Grey: neither rule is met. Every ordering must hold across 26 conditions: the main 2,000 papers per area, twenty half samples of 1,000, and five further selections of 2,000. Original and centred representations use the same papers. Centring moves the origin to each model’s mean over the balanced 52,000 unit vectors, then rescales the resulting vectors to length one. A cutoff of 5% requires a difference greater than 5% of the two area values’ average in every condition; the 0 cutoff requires a nonzero difference. Table S20 reports all four cutoffs and whether the same models retain the same opposing area orderings.

shows a problem that repeatability alone cannot solve: a geometric signal may not describe the feature of research we want to study.

5 Discussion

5.1 What should be checked before using a map

A high overall agreement score leaves some practical questions open. The models can share the broad arrangement of areas while directing a reader towards different papers or giving opposite comparisons between areas. Suppose one model gives area A a larger angular spread than B, and the result repeats with other articles. That shows stability to article selection within this model. If another model persistently reverses the order, more selections with the first model do not resolve the disagreement. The claim needs to be checked by changing the model as well as the articles.

The concern follows earlier work on method-dependent science maps (Gläser et al., 2017; Boyack and Klavans, 2020) and complements assessments against external classifications and relationships (González-Márquez et al., 2024; Constantino et al., 2025). Combining structural and neighbour comparisons also has precedent (Caspari et al., 2024).

Interpretation also depends on what the measure describes. Larger angles mean that vectors point further apart; a larger effective linear dimension means that their variation is less concentrated in a few directions. Those are descriptions of the representation. Calling an area more thematically diverse or interdisciplinary would require a further link between those numerical properties and the research itself. Related work on embedding geometry cautions against assigning a broad meaning to a convenient measure (Rudman et al., 2022). The failed fragmentation check illustrates why: even a geometric difference that repeats can support the wrong explanation.

For someone using these maps to compare research areas, the practical consequence is to check the comparison itself. An average agreement score describes what the representations share overall. Repeating article selection asks whether the answer depends on the papers chosen. Following the same opposing models adds a further check: whether a stable answer within each model transfers across representations. Reporting the size of the difference and the processing choices makes clear which of these claims the evidence supports. External evidence is still needed to interpret a geometric contrast as a property of the research itself.

5.2 What the additional checks change

The additional checks narrow the explanations for the main findings. The shared arrangement of area centres is not tied to one set of papers, to choosing 128 rather than 256 or 512, or to including one particular area. The low neighbour overlap is also not largely removed by the recorded quality filter: after excluding flagged and duplicate records, two models still share about eight papers in a list of 25. These checks strengthen those observations within the tested collection, without establishing the accuracy of the shared relationships.

For input text, the repeated selections preserve an average result that hides different model behaviours. Giving titles alone changes fewer neighbours than switching models for some models, and more for others. Replacing mean pooling with the first or last contextual output in the four word-level models also reverses the overall ordering. The practical consequence is to specify both what text a model reads and how its outputs become a paper vector. A statement that text and model choice matter equally would go beyond these results.

For angular spread, the common reference point is another consequential choice. Moving it to the mean of the balanced article set and rescaling the vectors changes many answers about which area has the larger spread. Repeating article selection does not remove this dependence or establish one representation as correct. PR changes less, although alternative dimension formulas still fail to retain every comparison. A claim about spread therefore needs to identify the representation, the measure and the size of the difference.

The discipline checks examine one concrete difference: models agree less about the relationships among Medicine papers than among Energy papers under the main rule (Supplement S5). This does not mean that Medicine is less coherent, and it does not always have the lowest neighbour overlap. The structural difference remains when the contributions of specialties within each area are balanced, and when the two biomedical models, BioBERT and PubMedBERT, are excluded. Neither tested explanation accounts for the whole difference. Models with related training also sometimes give similar results, but these apparent families change with the rule used to form paper vectors. With ten models that differ simultaneously in training texts, internal design and the task used to train them, these associations cannot identify one of those features as the cause.

The time comparison applies the same current models to papers from older and newer publication periods. Structural agreement increases slightly on average. Neighbour agreement falls by 0.87 percentage points when each group offers all its available papers, but rises by 0.91 points when every group offers 2,048 candidates. The apparent trend therefore depends on whether older and newer groups are allowed different search sizes. It describes agreement among present-day representations of those papers; it does not, by itself, show that science has historically converged.

5.3 What remains open

The collection requires English-labelled records with usable abstracts. Its reinforcement of smaller area-period groups helps comparison but changes proportions. Source errors remain, and cleaning is not a certification of every record. Repeated selections cannot repair missing coverage or make these results representative of all scientific work.

The ten models also set a boundary. Four were not trained specifically for document similarity. The six-model check shows that disagreement is not confined to those four, but does not make this set representative of all embedding models. To decide which area ordering

or article relationship is better supported, we need evidence outside these vectors, such as independent assessments of the papers.

Several extensions and controls also followed earlier results. Recording their rules before the new calculations makes the decisions traceable, but does not turn them into predictions made before the study began. A useful next step would test a specific map-based claim against independent judgments in a research area. Adding models alone would not settle what their shared or conflicting geometry means.

6 Conclusion

Changing the embedding model can preserve the broad arrangement of science while changing which papers appear nearby and which area has more widely spread representations. Giving a model titles alone can also produce large changes, with effects that depend on the model and the rule used to form paper vectors. Choosing other articles tests one source of dependence; changing models or the reference point tests others. Before using a map to support a particular relationship, we should check that relationship under the choices that can affect it and show how large the changes are. The tested models' agreement tells us what their representations share. Deciding whether that relationship describes the research well still needs independent evidence.

Declarations

Author contributions. Alejandro Treny Ortega: Conceptualization; Methodology; Software; Validation; Formal analysis; Investigation; Data curation; Visualization; Writing – original draft; Writing – review & editing.

Competing interests. The author declares no competing interests.

Funding. This research received no external funding.

Data and code availability. Code and analysis data (version 1.1.0) are publicly available on Zenodo at <https://doi.org/10.5281/zenodo.22876602> (Treny Ortega, 2026). A GitHub mirror provides the same files and commands. They preserve article identifiers, exact selections, measurements by model, counts by query and controls. The public route repeats the documented statistical comparisons from those frozen measurements; it excludes stored vectors, neighbour lists and model weights. Historical title/abstract tables are not redistributed because rights to the plaintext abstracts have not been established. Fresh OpenAlex retrieval may return different text, so an exact rerun from the historical inputs is not guaranteed. Code is MIT; original results and documentation are CC BY 4.0; OpenAlex metadata retain CC0. Supplement S8 details the tested scope.

References

- Iz Beltagy, Kyle Lo, and Arman Cohan. SciBERT: A pretrained language model for scientific text. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan, editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3615–3620, Hong Kong, China, nov 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1371. URL <https://aclanthology.org/D19-1371/>.
- Kevin W. Boyack and Richard Klavans. A comparison of large-scale science models based on textual, direct citation and hybrid relatedness. *Quantitative Science Studies*, 1(4):1570–1585, 2020. ISSN 2641-3337. doi: 10.1162/qss_a_00085.
- Laura Caspari, Kanishka Ghosh Dastidar, Saber Zerhoubi, Jelena Mitrovic, and Michael Granitzer. Beyond benchmarks: Evaluating embedding model similarity for retrieval augmented generation systems. arXiv:2407.08275, 2024. URL <https://arxiv.org/abs/2407.08275>. Preprint.
- Arman Cohan, Sergey Feldman, Iz Beltagy, Doug Downey, and Daniel Weld. SPECTER: Document-level representation learning using citation-informed transformers. In Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault, editors, *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 2270–2282, Online, jul 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.207. URL <https://aclanthology.org/2020.acl-main.207/>.
- Isabel Constantino, Sadamori Kojaku, Santo Fortunato, and Yong-Yeol Ahn. Representing the disciplinary structure of physics: A comparative evaluation of graph and text embedding methods. *Quantitative Science Studies*, 6:263–280, 2025. ISSN 2641-3337. doi: 10.1162/qss_a_00349.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In Jill Burstein, Christy Doran, and Tamar Solorio, editors, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota, jun 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1423. URL <https://aclanthology.org/N19-1423/>.
- Tianyu Gao, Xingcheng Yao, and Danqi Chen. SimCSE: Simple contrastive learning of sentence embeddings. In Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih, editors, *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 6894–6910, Online and Punta Cana, Dominican Republic, nov 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.emnlp-main.552. URL <https://aclanthology.org/2021.emnlp-main.552/>.
- Jochen Gläser, Wolfgang Glänzel, and Andrea Scharnhorst. Same data—different results? towards a comparative approach to the identification of thematic structures in science. *Scientometrics*, 111(2):981–998, 2017. ISSN 0138-9130. doi: 10.1007/s11192-017-2296-z.
- Rita González-Márquez, Luca Schmidt, Benjamin M. Schmidt, Philipp Berens, and Dmitry Kobak. The landscape of biomedical research. *Patterns*, 5(6):100968, 2024. ISSN 2666-3899. doi: 10.1016/j.patter.2024.100968.
- Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare*, 3(1):1–23, 2022. ISSN 2691-1957. doi: 10.1145/3458754.

- Matthias Held and Theresa Velden. How to interpret algorithmically constructed topical structures of scientific fields? a case study of citation-based mappings of the research specialty of invasion biology. *Quantitative Science Studies*, 3(3):651–671, 2022. ISSN 2641-3337. doi: 10.1162/qss__a__00194.
- Nathaniel Imel and Zachary Hafen. Density, asymmetry and citation dynamics in scientific literature. arXiv:2506.23366, 2025. URL <https://arxiv.org/abs/2506.23366>. Preprint.
- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 3519–3529. PMLR, 09–15 Jun 2019. URL <https://proceedings.mlr.press/v97/kornblith19a.html>.
- Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo Kang. Biobert: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4):1234–1240, 2020. ISSN 1367-4803. doi: 10.1093/bioinformatics/btz682.
- Zhentao Liang, Nees Jan van Eck, Xuehua Wu, Jin Mao, and Gang Li. Citation importance-aware document representation learning for large-scale science mapping. *Information Processing & Management*, 63(3):104557, 2026. ISSN 0306-4573. doi: 10.1016/j.ipm.2025.104557.
- Alex Murphy, Joel Zylberberg, and Alona Fyshe. Correcting biased centered kernel alignment measures in biological and artificial neural networks. arXiv:2405.01012, 2024. URL <https://arxiv.org/abs/2405.01012>. Preprint.
- OpenAlex. Topics. OpenAlex Help Center, 2026. URL <https://help.openalex.org/data/topics/>. Accessed 18 September 2026.
- Malte Ostendorff, Nils Rethmeier, Isabelle Augenstein, Bela Gipp, and Georg Rehm. Neighborhood contrastive learning for scientific document representations with citation embeddings. In Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang, editors, *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 11670–11688, Abu Dhabi, United Arab Emirates, dec 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-main.802. URL <https://aclanthology.org/2022.emnlp-main.802/>.
- Jason Priem, Heather Piwowar, and Richard Orr. OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts, 2022. URL <https://arxiv.org/abs/2205.01833v2>. Preprint, version 2.
- Prashant C. Raju. Geometric stability: The missing axis of representations. arXiv:2601.09173, 2026. URL <https://arxiv.org/abs/2601.09173>. Preprint.
- Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan, editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, Hong Kong, China, nov 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1410. URL <https://aclanthology.org/D19-1410/>.

- William Rudman, Nate Gillman, Taylor Rayne, and Carsten Eickhoff. IsoScore: Measuring the uniformity of embedding space utilization. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio, editors, *Findings of the Association for Computational Linguistics: ACL 2022*, pages 3325–3339, Dublin, Ireland, may 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.findings-acl.262. URL <https://aclanthology.org/2022.findings-acl.262/>.
- Amanpreet Singh, Mike D’Arcy, Arman Cohan, Doug Downey, and Sergey Feldman. SciRepEval: A multi-format benchmark for scientific document representations. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 5548–5566, Singapore, dec 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.338. URL <https://aclanthology.org/2023.emnlp-main.338/>.
- Shruti Singh and Mayank Singh. The inefficiency of language models in scholarly retrieval: An experimental walk-through. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio, editors, *Findings of the Association for Computational Linguistics: ACL 2022*, pages 3153–3173, Dublin, Ireland, may 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.findings-acl.249. URL <https://aclanthology.org/2022.findings-acl.249/>.
- Alejandro Treny Ortega. How much does the map of science depend on the embedding model? compact reproducibility data and code. Zenodo, version 1.1.0, data set, 2026. URL <https://doi.org/10.5281/zenodo.22876602>.
- Alex H Williams. Equivalence between representational similarity analysis, centered kernel alignment, and canonical correlations analysis. In Marco Fumero, Clementine Domine, Zorah Lähner, Donato Crisostomi, Luca Moschella, and Kimberly Stachenfeld, editors, *Proceedings of UniReps: the Second Edition of the Workshop on Unifying Representations in Neural Models*, volume 285 of *Proceedings of Machine Learning Research*, pages 10–23. PMLR, 14 Dec 2024. URL <https://proceedings.mlr.press/v285/williams24a.html>.
- Alex H Williams, Erin Kunz, Simon Kornblith, and Scott Linderman. Generalized shape metrics on neural representations. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, volume 34, pages 4738–4750. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/252a3dbaeb32e7690242ad3b556e626b-Paper.pdf.